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Project Charter

Supply Chain Performance & Analytics Dashboard

Document Version History

Version	Date	Author	Description
1.0	25-Aug-2026	Shahadat Hossain Sajim	Initial Draft
1.1	05-Sep-2026	Shahadat Hossain Sajim	Final Portfolio Version

Project Information

Field	Value
Project Name	Supply Chain Performance & Analytics Dashboard
Project Type	Data Analytics & Business Intelligence
Industry	Supply Chain & Retail
Role	Data Analyst
Document Status	Final
Confidentiality	Portfolio Project
Prepared By	Shahadat Hossain Sajim
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1. Project Overview

This project analyzes end-to-end supply chain performance across procurement, inventory, production, and sales. The analysis uses SQL to explore operational data hosted on Supabase and Power BI to build an interactive dashboard for monitoring key performance indicators and identifying business insights.

2. Problem Statement

Supply chain data is often spread across different operational areas, making it difficult to understand overall performance and identify potential issues.

This project aims to bring key supply chain metrics together and answer practical business questions related to:

- Supplier procurement performance
- Inventory availability and risk
- Production output and quality
- Product sales and profitability

3. Project Objectives

The main objectives of this project are to:

- Analyze supply chain data using SQL.
- Evaluate procurement, inventory, production, and sales performance.
- Identify inventory risks and production quality issues.
- Understand product-level sales and profitability.
- Build an interactive Power BI dashboard for business monitoring.
- Turn operational data into clear and actionable insights.

4. Project Scope (In Scope / Out of Scope)

In Scope

The project covers:

- **Procurement:** supplier spending, procurement quantity, cost, lead time, and quality.
- **Inventory:** current stock, reorder-point risk, inventory shortfalls, and estimated inventory value.
- **Production:** production output and defective units.
- **Sales:** revenue, units sold, profit, profit margin, product performance, and revenue trends.

Out Of Scope

- The shipment data was reviewed during data validation but was not included in the final integrated data model because its key IDs were not consistent with the available product, customer, and facility dimensions.
- Capacity utilization analysis was also excluded because the available production and capacity data were not sufficiently aligned for a reliable comparison.

5. Key Business Questions

The analysis focuses on questions such as:

- Which suppliers account for the highest procurement spend?
- Which products and suppliers have higher procurement costs or lead times?
- Which products and facilities currently face inventory risk?
- Which facilities produce the highest output?
- Which products have the highest defective units?
- Which products generate the highest revenue and profit?
- Which products have the highest profit margins?

6. Data Sources

The project uses CSV datasets covering:

Dimension Tables

- dim_product
- dim_customer
- dim_facility
- dim_supplier
- dim_date

Fact Tables

- fact_procurement
- fact_inventory
- fact_production
- fact_sales

The datasets are hosted in Supabase and queried using SQL

7. Tools & Technologies

Tool	Purpose
Supabase / PostgreSQL	Data hosting and SQL analysis
SQL	Data exploration and business analysis
Power BI	Dashboard development and visualization
DAX	KPI and analytical measures
GitHub	Project documentation and version control

8. Key Deliverables

The project delivers:

- SQL-based supply chain analysis
- Data validation and quality checks
- Integrated Power BI data model
- 4-page interactive Power BI dashboard
- KPI and DAX measures
- Business insights and recommendations
- GitHub project documentation

9. Dashboard Structure

The Power BI dashboard contains four analytical pages:

Page 1 — Executive Overview

Provides a high-level view of revenue, units sold, profit, production, and major

product/facility performance.

Page 2 — Procurement & Supplier

Focuses on procurement spending, quantity, supplier performance, procurement cost, and lead time.

Page 3 — Inventory & Production Performance

Covers inventory risk, current inventory, production output, defective units, and estimated inventory value.

Page 4 — Sales & Product Performance

Analyzes product profitability, sales volume, product-line performance, and revenue by category.

10. Success Criteria

The project is considered successful when it:

- Provides reliable and validated supply chain analysis.
- Clearly connects procurement, inventory, production, and sales data.
- Highlights important operational risks and performance trends.
- Provides an interactive and easy-to-understand Power BI dashboard.
- Enables users to explore performance using relevant date, product, facility, supplier, and category filters.
- Converts raw operational data into meaningful business insights.

11. Project Workflow

Data → Supabase → SQL Analysis → Data Validation → Power BI Data Model → DAX Measures → Dashboard → Business Insights